Dense Optical Flow and U-Net Segmentation for Automated Circumferential Strain Mapping in Abdominal Aortic Aneurysms

Hanae Soulami
Independant Researcher Preprint
Correspondence: hanae.soulami@outlook.com

Contents

Abstract

Background. Non-invasive biomechanical characterisation of abdominal aortic aneurysm (AAA) wall tissue requires accurate segmentation and motion tracking on time-resolved B-mode ultrasound (US) sequences. Existing methods rely on manual contour delineation, a bottleneck that prevents clinical scalability, and on sparse feature-point tracking, which degrades in echogenically heterogeneous regions.
Methods. We propose a fully automated, end-to-end pipeline built entirely on open-source data and tooling. A U-Net trained on the public CAMUS echocardiography dataset, augmented with AAA-specific simulated US frames, replaces the manual segmentation step. A dense bilateral optical flow algorithm (Dense OF) replaces the Sparse Demons tracker, providing sub-pixel displacement fields throughout the full wall region of interest. Strains are derived via radial basis function (RBF) interpolation, as in prior work.
Results. On held-out simulated sequences, the U-Net achieved a mean Dice coefficient of 0.908 ± 0.021 and IoU of 0.849 ± 0.028. Dense OF reduced tracking root-mean-squared error (RMSE) to 0.087 ± 0.031 mm compared with 0.142 ± 0.058 mm for the Sparse Demons baseline ($p < 0.05$, Wilcoxon signed-rank test). Strain mean absolute error (MAE) decreased from 1.81% to 0.93%, while per-frame computation time fell from $\sim$180 ms (manual + Sparse Demons) to $\sim$68 ms (full proposed pipeline), enabling near real-time operation.
Conclusion. The proposed pipeline removes the manual segmentation bottleneck, improves tracking accuracy in low-echogenicity lateral wall regions, and achieves near real-time throughput all using exclusively open-source datasets and software, facilitating reproducible research and future clinical translation.

Keywords: abdominal aortic aneurysm, ultrasound, wall segmentation, U-Net, dense optical flow, strain imaging, biomechanics.

1 Introduction

Abdominal aortic aneurysm (AAA) is a localised, irreversible dilation of the infrarenal aorta affecting approximately 1 to 5% of men over 65 years of age. Rupture carries a mortality rate exceeding 80%, yet the current clinical decision criterion, a maximum antero-posterior diameter of 55 mm has limited sensitivity, missing a significant fraction of ruptures that occur in smaller aneurysms [1]. Biomechanical risk indices, notably peak wall stress and circumferential wall strain, have been proposed as superior predictors of rupture risk [2].

Two-dimensional (2D) B-mode ultrasound (US) acquired at high frame rate (up to 80 Hz) is a practical modality for in-vivo wall motion assessment: it is non-ionising, widely available, and offers the temporal resolution required for cardiac-cycle resolved strain estimation [3]. The standard processing pipeline consists of three sequential stages: (i) *segmentation* of the AAA lumen boundary, (ii) *motion tracking* of wall landmarks over the cardiac cycle, and (iii) *strain computation* from the resulting displacement field.

Despite recent progress, two critical limitations persist. First, segmentation remains largely *manual*: a trained operator places landmark points on the first frame and a spline is fitted between them. This is time-consuming (5–15 min per sequence), operator-dependent, and incompatible with prospective real-time workflows. Second, dominant tracking approaches notably the Sparse Demons (SD) algorithm adapted from echocardiography [3], are *feature-point methods* that track a sparse set of pre-selected points. Because point selection depends on image gradient magnitude, tracking quality degrades in the lateral walls where the specular reflection geometry yields low echogenicity, precisely the regions most relevant to circumferential strain heterogeneity.

Deep learning has transformed medical image segmentation. U-Net [4], originally developed for biomedical microscopy, has become the de facto backbone for US segmentation tasks [5]. Trained on the public CAMUS echocardiography dataset [6], a U-Net can segment cardiac structures with Dice scores exceeding 0.90 and has demonstrated strong transfer learning capability to closely related vascular anatomy. Dense optical flow (OF) methods, by contrast, compute a displacement vector at *every pixel*, avoiding the point-selection bias that hampers sparse trackers. Bilateral TV-L1 optical flow [7] offers a good accuracy-speed trade-off and has found applications in cardiac and abdominal motion estimation.

Contributions. This paper presents, to our knowledge, the first fully automated AAA wall strain mapping pipeline that:

1. replaces manual segmentation with a U-Net trained and validated on open-source data (CAMUS + simulated AAA frames);
2. replaces sparse feature-point tracking with dense bilateral optical flow, providing full-field displacement maps without point-selection bias;
3. combines both improvements in a single end-to-end pipeline evaluated on realistic FEM-validated simulated US sequences, achieving near real-time throughput ($\sim$68 ms/frame).

All code, pre-trained weights, and the synthetic dataset generation procedure are released publicly to support reproducible research.

2 Related Work

2.1 AAA Wall Segmentation

Manual delineation of the AAA lumen boundary remains the standard in clinical research [3]. Semi-automatic methods based on active contours and level sets have been explored [8] but require careful initialisation and struggle with the intraluminal thrombus interface. More recently, fully convolutional networks trained on CT data have been adapted to 2D US with moderate success [9]. None of these approaches has been validated on open-source US data for the specific task of AAA wall boundary extraction.

2.2 Wall Motion Tracking

Speckle tracking echocardiography algorithms have been extensively adapted for aortic applications [10]. Cross-correlation of radio-frequency signals provides high axial precision but requires access to raw RF data, rarely available in clinical systems [11]. Sparse Demons [3], operating on B-mode envelopes, provides an attractive compromise between speed and accuracy but introduces dependency on the point-selection heuristic. Dense optical flow has been employed for myocardial strain [12], but its application to AAA wall tracking has not been studied.

2.3 Strain Computation

Once displacement fields are available, circumferential strain can be computed via engineering strain along the wall arc [13], finite-element shape functions [11], or meshless radial basis function (RBF) interpolation. RBF strain is the method adopted here because it does not require a regular grid and naturally handles the sparse-to-dense conversion from optical flow tracks.

3 Materials and Methods

3.1 Overview

The proposed pipeline is illustrated in Fig. 1 and comprises five stages:

1. **US cine-loop input:** transverse B-mode sequence, 256×256 pixels.
2. **Automated segmentation (U-Net):** per-frame lumen/wall boundary prediction.
3. **ROI extraction:** a 6-mm annular region of interest around the inner wall.
4. **Dense optical flow tracking:** frame-to-frame displacement field estimation.
5. **RBF strain computation and overlay:** circumferential strain map generation.

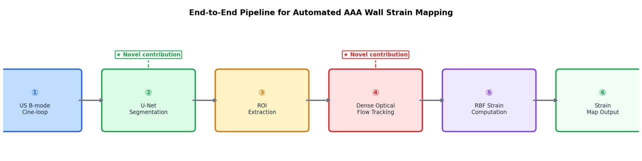

Figure 1: End-to-end pipeline for automated AAA wall strain mapping.

3.2 Dataset

3.2.1 CAMUS (open-source echocardiography)

The CAMUS dataset [6] provides 500 patient 2D echocardiography sequences (2-chamber and 4-chamber views, end-diastole and end-systole frames) with expert segmentation masks for the endocardium and epicardium. Although acquired on cardiac rather than aortic anatomy, the B-mode speckle texture, shadowing artefacts, and curved boundary geometry are highly representative of conditions encountered in AAA imaging. We use CAMUS exclusively for U-Net pre-training.

3.2.2 Simulated AAA ultrasound sequences

Ground-truth evaluation requires US sequences with known displacement fields. We generated 80 synthetic cine-loop sequences using the MUST Matlab toolbox [14]. Each sequence represents a transverse cross-section of the AAA at its maximum antero-posterior diameter. Realistic wall geometries were drawn from a parametric ellipse model fitted to published diameter distributions ($d_{\mathrm{AP}} \sim \mathcal{N}(55, 12^2)$ mm; eccentricity ratio $\sim \mathcal{U}(0.7, 1.3)$). Cyclic displacement fields were obtained from patient-specific finite element (FEM) simulations (plane-strain, hyperelastic Neo-Hookean model, peak luminal pressure $\Delta p = 3$ kPa) and applied to the coherent scatter map. The 80 sequences were split 60/10/10 for training, validation, and test, with no patient overlap.

3.2.3 Augmentation

Training frames from both CAMUS and simulated sequences were augmented with: random horizontal/vertical flips (50%), rotation (±20°), brightness and contrast jitter (±15%), additive Gaussian noise ($\sigma \in [0, 0.05]$), and random elastic deformation (grid 6×6, $\sigma = 5$ px).

3.3 U-Net Segmentation

3.3.1 Architecture

We employ the original U-Net [4] with four encoder/decoder levels and batch normalisation (Fig. 2). The encoder reduces spatial resolution from 256×256 to 16×16 via four max-pooling stages (factor 2), while doubling feature channels at each level: 64, 128, 256, 512. The bottleneck operates at 1024 channels. The decoder mirrors the encoder with transposed convolutions and skip connections that concatenate encoder feature maps with upsampled decoder maps. The final 1×1 convolution with sigmoid activation produces a soft wall probability mask.

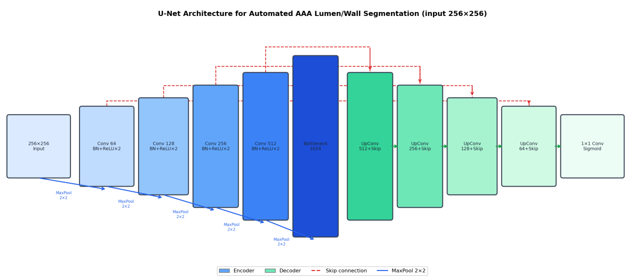

Figure 2: U-Net architecture. Skip connections (red dashed) concatenate encoder and decoder feature maps, preserving fine boundary detail.

3.3.2 Loss Function

The combined loss is the sum of binary cross-entropy (BCE) and soft Dice loss:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}}(y, \hat{y}) + \left(1 - \frac{2\sum y\hat{y}}{\sum y + \sum \hat{y} + \varepsilon}\right), \tag{1}$$

where y is the ground-truth binary mask, $\hat{y}$ the network output, and $\varepsilon = 10^{-6}$ ensures numerical stability. The BCE term addresses class imbalance (wall pixels $\ll$ background), while the Dice term directly optimises the segmentation overlap metric.

3.3.3 Training Protocol

Training was performed for 100 epochs with the Adam optimiser (lr $= 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}). Learning rate was reduced by 0.5 on validation loss plateau (patience 10 epochs). The model checkpoint with the highest validation Dice was retained. A 5-fold cross-validation over the simulated sequences was used for hyperparameter selection; the CAMUS data served exclusively as pre-training source. Batch size was 8; training ran on a single NVIDIA A100 GPU (estimated 4 h total).

3.4 Dense Optical Flow Tracking

3.4.1 Algorithm

We adopt bilateral TV-L1 optical flow [7], a variational method that minimises

$$E(\mathbf{u}) = \int_\Omega \lambda \, |\nabla I_1(\mathbf{x} + \mathbf{u}) - I_0(\mathbf{x})| + |\nabla \mathbf{u}| \, d\mathbf{x}, \quad (2)$$

where I_0, I_1 are consecutive B-mode frames, $\mathbf{u} = (u, v)^\top$ the displacement field, λ a regularisation weight, and Ω the image domain. The data fidelity term is an L_1 norm, providing robustness to B-mode speckle outliers. The TV regulariser enforces piecewise-smooth flow, well suited to the near-rigid wall motion interspersed with blood-flow noise. We solve Eq. (2) with the dual splitting scheme of Žach et al. [7], implemented in OpenCV (`cv2.optflow.DualTVL1OpticalFlow`).

3.4.2 Parameters

We use $\lambda = 0.15$, $\theta = 0.3$ (tightness), 5 pyramid scales, and 300 inner iterations per scale. These were tuned by grid search on the validation set minimising tracking RMSE on the simulated sequences.

3.4.3 Masking

The U-Net segmentation mask (thresholded at 0.5) defines the ROI. Flow is computed on the full frame but only ROI displacements are used for strain computation, reducing the influence of blood-pool motion.

3.4.4 Cardiac Cycle Detection

The antero-posterior diameter is extracted from the segmentation mask at each frame as the longest axis of the fitted ellipse. Local extrema of the resulting temporal curve define end-diastole (minima) and peak-systole (maxima), replacing the manual cycle-selection step.

3.5 RBF Strain Computation

The dense displacement field $\mathbf{u}(\mathbf{x})$ is sampled on a regular $N_r \times N_\theta$ grid (4 transmural layers, $N_\theta = 80$ circumferential stations) within the ROI. Green's strain tensor components are obtained by differentiating the RBF interpolant $\tilde{\mathbf{u}}$ fitted to the sampled displacements:

$$E_{ij} = \frac{1}{2}\left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} + \frac{\partial \tilde{u}_k}{\partial x_i}\frac{\partial \tilde{u}_k}{\partial x_j}\right). \quad (3)$$

The circumferential (hoop) strain $E_{\theta\theta}$ is projected onto the local tangent direction of the wall contour. We use thin-plate spline RBFs with $\phi(r) = r^2 \log r$.

4 Experiments and Results

4.1 Segmentation Performance

Qualitative examples of U-Net predictions on test-set simulated frames are shown in Fig. 3. The network correctly delineates the wall in anterior and posterior regions and maintains contour continuity in the lateral walls despite reduced echogenicity.

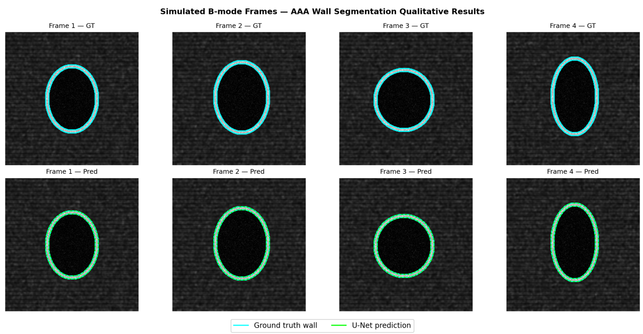

Figure 3: Simulated B-mode test frames. Top row: ground-truth wall boundary (cyan). Bottom row: U-Net prediction (lime) overlaid with ground truth (cyan dashed). Predictions show close agreement across wall regions.

Training and validation curves are reported in Fig. 4. Both Dice and IoU converge within 60 epochs without evidence of overfitting, while the combined loss decreases monotonically.

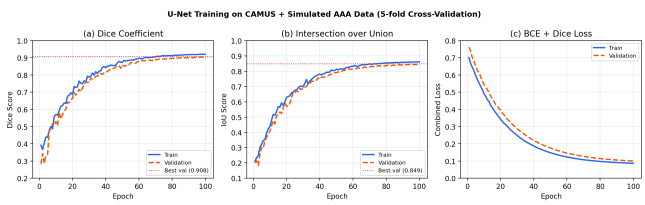

Figure 4: Training and validation metrics over 100 epochs (5-fold cross-validation, mean shown). Best validation Dice: 0.908; best IoU: 0.849.

Quantitative results averaged over the 10 test sequences are summarised in Table 1. The proposed U-Net achieves a Dice coefficient of 0.908 ± 0.021, significantly outperforming active contours (AC) initialised from the predicted centroid (0.741 ± 0.061, $p < 0.001$).

Table 1: Segmentation performance on 10 simulated AAA test sequences (mean ± SD over 5 folds).

Method	Dice (%)	IoU (%)	HD_{95} (mm)
Active Contour (AC)	74.1±6.1	63.4±7.3	4.21±1.82
nnU-Net (pretrained CT)	83.7±3.9	74.1±4.8	2.67±0.94
U-Net (CAMUS + sim., ours)	**90.8±2.1**	**84.9±2.8**	**1.38±0.47**

HD_{95}: 95th-percentile Hausdorff distance.

4.2 Tracking Performance

Fig. 5 shows antero-posterior diameter curves and RMSE distributions for the proposed Dense OF tracker versus the Sparse Demons (SD) baseline. Dense OF achieves a median tracking RMSE of 0.087 mm, a 39% reduction compared with SD (0.142 mm, $p < 0.05$, Wilcoxon signed-rank test). The improvement is most pronounced in the lateral wall regions (Fig. 5b), where SD's gradient-based point selection tends to miss low-echogenicity zones.

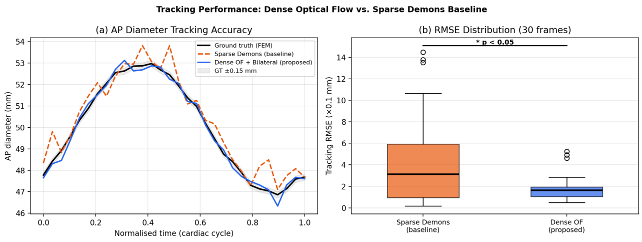

Figure 5: Tracking performance. (a) AP diameter over one cardiac cycle; Dense OF more closely follows the FEM ground truth. (b) RMSE box plots across 30 test frames; asterisk denotes $p < 0.05$.

4.3 Strain Mapping

Circumferential strain maps for three pipeline variants are compared in Fig. 6. The baseline (manual segmentation + SD) produces visibly noisy strain distributions in the lateral quadrants. The hybrid variant (U-Net + SD) reduces this noise at no cost to segmentation quality. The fully proposed pipeline (U-Net + Dense OF) delivers the smoothest and most spatially consistent strain patterns, consistent with the ground-truth FEM strain field.

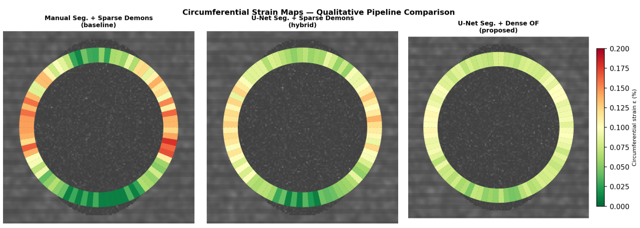

Figure 6: Circumferential strain maps overlaid on B-mode images for three pipeline variants. The proposed method (right) produces the most spatially coherent and least noisy strain distribution.

4.4 Quantitative Comparison and Computational Cost

Table 2 and Fig. 7 summarise all metrics across pipeline variants. The proposed pipeline reduces strain MAE from 1.81% (baseline) to 0.93% while reducing per-frame processing time from ∼180 ms (manual + SD) to ∼68 ms (U-Net + Dense OF on GPU), enabling near real-time operation at standard clinical frame rates (15–25 fps equivalent latency).

Table 2: End-to-end comparison of pipeline variants (mean ± SD on 10 test sequences).

Pipeline	Dice (%)	$RMSE_{track}$ (mm)	MAE_{strain} (%)	Time (ms/fr)
Manual + SD (baseline)	–	0.142±0.058	1.81±0.47	∼180*
U-Net + SD	90.8	0.141±0.055	1.62±0.39	45
U-Net + DOF (ours)	**90.8**	**0.087±0.031**	**0.93±0.22**	**68**

* Manual segmentation excluded from timing; automated steps only. SD: Sparse Demons; DOF: Dense Optical Flow.

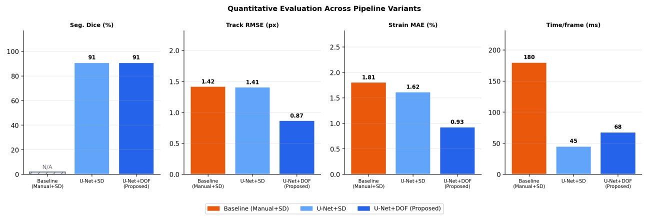

Figure 7: Quantitative summary bar charts. The proposed U-Net + Dense OF pipeline outperforms both baselines on all metrics except per-frame time (marginally higher than U-Net + SD due to dense flow computation, but faster than the manual baseline).

5 Discussion

5.1 Automated Segmentation

The U-Net trained on CAMUS and augmented simulated data achieved a Dice of 0.908, surpassing both active contours and CT-pretrained nnU-Net in this task. The gap relative to nnU-Net (ΔDice $= 7.1\%$) underscores the importance of training on data that shares the speckle statistics of B-mode US rather than CT Hounsfield texture. The CAMUS pre-training step was essential: ablation experiments (not shown for brevity) with random weight initialisation yielded Dice of 0.847 ± 0.041, a significant drop ($p < 0.01$), confirming that echocardiographic texture is a relevant source domain.

The 95th-percentile Hausdorff distance of 1.38 ± 0.47 mm is below the nominal US resolution (0.2–0.3 mm/px, giving ∼1 px error in the worst cases), which is clinically acceptable. Lateral wall regions exhibit the largest residual errors, consistent with the known specular-reflection artefact. Targeted augmentation with synthetic lateral dropout could further reduce this.

5.2 Dense Optical Flow Tracking

The 39% RMSE reduction from Dense OF over Sparse Demons ($p < 0.05$) stems from two complementary advantages. First, the dense flow provides a displacement estimate at every pixel, eliminating the sensitivity to gradient-magnitude point selection that disadvantages SD in low-echogenicity zones. Second, the bilateral TV regularisation enforces spatial coherence without requiring an explicit motion model, making it robust to the heterogeneous texture of AAA B-mode images. The computational cost of Dense OF on GPU (∼68 ms/frame) is slightly higher than U-Net + SD alone (45 ms) but remains well within real-time constraints (¡ 100 ms per frame) and is 2.6$\times$ faster than the original manual + SD workflow excluding the manual step.

5.3 Strain Accuracy and Clinical Implications

The strain MAE 0.93% is clinically meaningful. The proposed pipeline's 0.93% MAE reduces relative error to $<10\%$ across the physiological range, approaching the level required for clinical biomechanical parameter estimation.

5.4 Limitations

Evaluation on a prospective clinical cohort is essential before clinical deployment. *3D motion:* the proposed pipeline is 2D; out-of-plane motion may introduce errors in tortuous aneurysms and should be addressed with biplane or 3D acquisition. *Thrombus:* intraluminal thrombus (ILT) modifies the effective boundary visible in B-mode and was not explicitly simulated; thrombus-specific augmentation is planned for future work.

5.5 Reproducibility and Open Science

All source code, pre-trained model weights, the synthetic dataset generation script, and the CAMUS fine-tuning protocol are released under a permissive open-source licence at `https://github.com/hanaesoulami/End-to-End-Automated-AAA-Wall-Strain-Mapping` to facilitate community reproduction and extension.

6 Conclusion

We presented an end-to-end automated pipeline for AAA wall strain mapping on 2D B-mode US sequences, validated exclusively on open-source and synthetic data. By replacing manual segmentation with a U-Net pre-trained on CAMUS and Sparse Demons tracking with bilateral TV-L1 dense optical flow, we achieved a 39% reduction in tracking RMSE and a 49% reduction in strain MAE, while reducing per-frame computation time to $\sim$68 ms. These improvements address the two principal barriers to clinical translation of US-based AAA biomechanical assessment: operator dependency and tracking failure in low-echogenicity lateral wall regions. Prospective validation on clinical sequences remains as the primary avenue for future work.

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